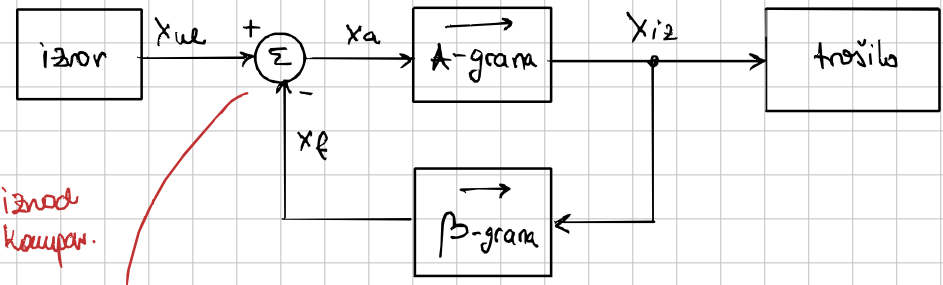
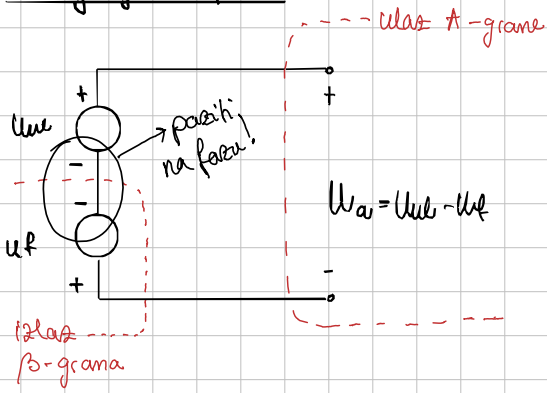
Struktura pojačala s povratnom vezom

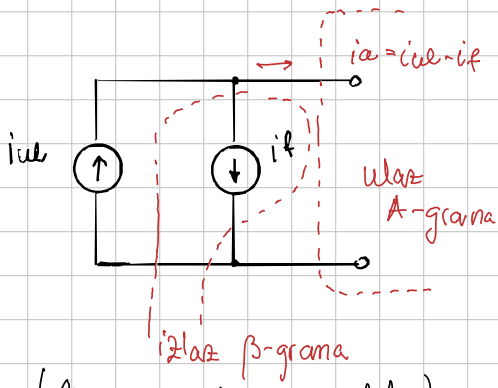


izvod kompon.

Zbrajanje napona:



Zbrajanje struja:



I. TIP PV

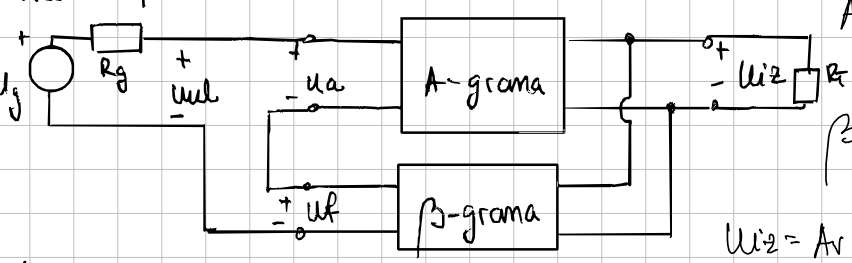
X_{iz} = napon – voltmetar (β-grana spajamo paralelno)

X_f = napon – spajamo u seriji

⟨veličina koju mjerimo⟩ – ⟨tip zbrajanja u ulaznom krugu⟩

NAPONSKA – SERIJSKA P.V

X_a = napon



$$A_v = \frac{U_{iz}}{U_a}$$

$$\beta = \frac{U_f}{U_{iz}}$$

$$U_{iz} = A_v \cdot U_a = A_v(U_{ul} - U_f)$$

$$U_a = U_{ul} - U_f$$

$$U_{iz} = A_v(U_{ul} - \beta U_{iz}$$

$$U_{iz}(1 + \beta A_v) = A_v \cdot U_{ul}$$

$$\frac{U_{iz}}{U_{ul}} = A_{vf} = \frac{A_v}{1 + \beta A_v}$$

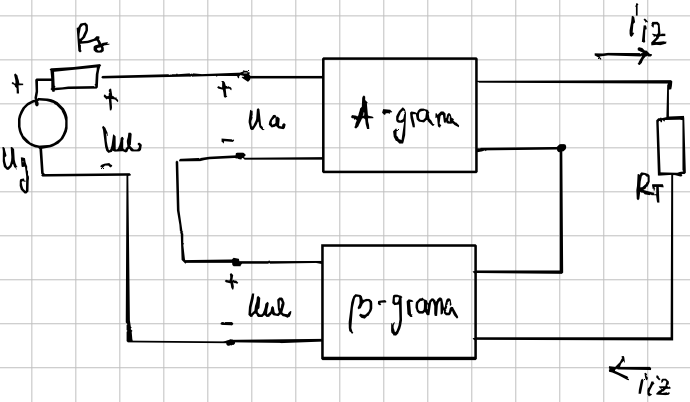
βA_v je uvijek bezdimenzionalna veličina!

II. TIP PV

X_{iz} = STRUJA – β-grana spojena kao AMPER. (serija)

X_f = NAPON – zbrajamo u seriji

STRUJNA – SERIJSKA P.V



$$G_M = \frac{i_{iz}}{U_a} \quad [S]$$

$$\beta = \frac{U_f}{i_{iz}} \quad [\Omega]$$

βG_M – bezdimenzionalan

$$i_{iz} = G_M \cdot U_a = G_M (U_{ul} - U_f)$$

$$i_{iz} = G_M (U_{ul} - i_{iz} \cdot \beta)$$

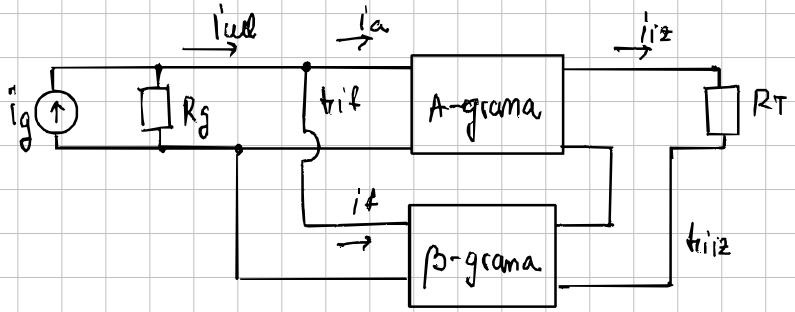
$$\frac{i_{iz}}{U_{ul}} = G_{Mf} = \frac{G_M}{1+\beta G_M}$$

III. TIP PV

X_{iz} = STRUJA → serijski (β-grana)

X_f = STRUJA – zbrajamo u paraleli

STRUJNA – PARALELNA PV



$$i_a = i_{ul} - i_f$$

$$A_I = \frac{i_{iz}}{i_a} \quad [\ ]$$

$$\beta = \frac{i_f}{i_{iz}} \quad [\ ]$$

βA_I – bezdimenz

$$i_{iz} = A_I \cdot i_a = A_I (i_{ul} - i_f) = A_I (i_{ul} - \beta \cdot i_{iz})$$

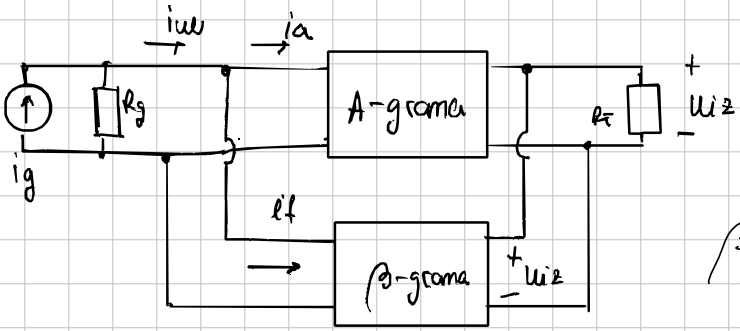
$$\frac{i_{iz}}{i_{ul}} = A_{If} = \frac{A_I}{1+\beta A_I}$$

IV. TIP

X_{iz} = napon — paralelno

X_f = struja — zbrajamo u paraleli

NAPONSKA - PARALELNA P.V



$$R_M = \frac{u_{iz}}{i_a} \quad [\Omega]$$

$$\beta = \frac{i_f}{u_{iz}} \quad [S]$$

βR_M — bezdimenz.

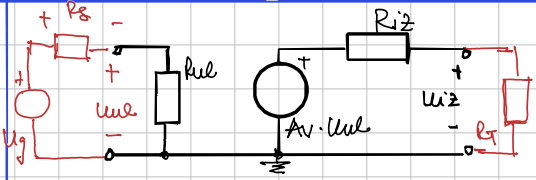
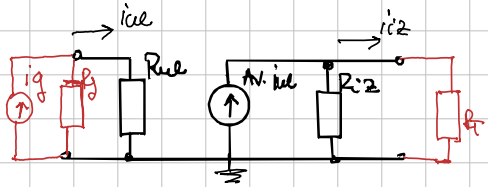
$$u_{iz} = R_M \cdot i_a = R_M (i_{ul} - i_f) = R_M (i_{ul} - \beta u_{iz})$$

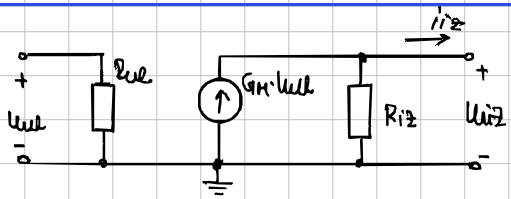
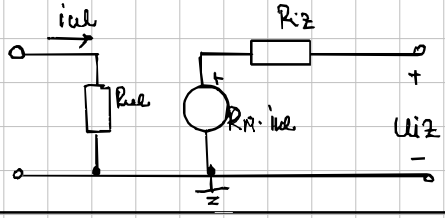
$$u_{iz}(1+\beta R_M) = R_M \cdot i_{ul}$$

$$\frac{u_{iz}}{u_{ul}} = R_{Mf} = \frac{R_M}{1+\beta R_M}$$

Kako povratna veza utječe na ulazne / izlazne otpore

Tipovi pojačala:

Tip	Model	Pojačanje	Idealno pojačalo
Naponsko pojačalo		$A_v = \left.\frac{U_{iz}}{U_{ul}}\right\|_{i_c=0}$	$R_{ul} \to \infty$ $R_{iz} \to 0$
Strujno pojačalo		$A_i = \left.\frac{i_{iz}}{i_{ul}}\right\|_{u_c=0}$	$R_{ul} \to 0$ $R_c \to \infty$

Tip	Model	Pojačanje	Idealno poj.
Strujno pojačalo		$G_M = \left.\frac{i_{iz}}{U_{ul}}\right\|_{u_c=0}$	$R_{ul} \to \infty$ $R_{iz} \to 0$
Otporno pojačalo		$R_M = \left.\frac{U_{iz}}{i_{ul}}\right\|_{i_c=0}$	$R_{ul} \to 0$ $R_{iz} \to 0$